

Chapter 7: Hypothesis Testing

A Level Statistics Notes · Rexponential

Why this chapter matters

Every previous chapter has described data or built a probability model. Hypothesis testing turns that machinery into a decision-making tool: given some observed data, is there enough evidence to conclude that something genuine has changed, or could the result plausibly have arisen by chance? This chapter applies the binomial and Normal distributions from Chapter 6, and the correlation coefficient from Chapter 4, to a single formal procedure that is reused with only minor variations across the rest of the course.

7.1 The language of hypothesis testing

Concept

A hypothesis test starts with a claim about a population parameter (for example, a probability or a mean) and uses sample data to decide whether that claim should be rejected. The **null hypothesis**, H_0 , is the value of the parameter that is assumed to be true unless the data give strong evidence otherwise. The **alternative hypothesis**, H_1 , states what would be true instead if the parameter has genuinely changed. H_1 can point in one specific direction (a one-tailed test) or allow either direction (a two-tailed test), depending on how the claim is worded.

To decide between H_0 and H_1 , the **p-value** — the probability of getting a result at least as extreme as the one observed, assuming H_0 is true — is calculated and compared with the **significance level**, a threshold probability chosen in advance (commonly 5% or 1%). If the p-value is smaller than the significance level, the result is judged too unlikely to have happened by chance alone, so H_0 is rejected. The set of sample outcomes that would lead to this rejection is called the **critical region**.

Key vocabulary

Term	Meaning
Null hypothesis, H_0	The value of the parameter assumed to be true unless the data give evidence otherwise
Alternative hypothesis, H_1	What would be true if the parameter has genuinely changed
Significance level	The threshold probability for rejecting H_0 (commonly 5% or 1%)
One-tailed / two-tailed test	Testing for a change in one specific direction, or in either direction
Critical region	The set of outcomes that would lead to rejecting H_0
p-value	The probability of a result at least as extreme as the one observed, assuming H_0 is true

Procedure

1. Define the parameter being tested in words and set up H_0 and H_1 in terms of it.

- Decide whether the test is one-tailed (H_1 uses $<$ or $>$) or two-tailed (H_1 uses $\neq$) based on the wording of the claim being tested.
- Either find the p-value of the observed result and compare it to the significance level, or find the critical region and check whether the observed result falls inside it.
- State a conclusion in context: whether there is sufficient evidence to reject H_0 , referring back to the original claim — never simply say “reject H_0 ” without relating it to the scenario.

Worked example

EXAMPLE 7.1

A manufacturer claims that 30% of customers prefer their new product design. A researcher wants to test, at the 5% significance level, whether this proportion has increased. Write down suitable null and alternative hypotheses.

Step 1. Let p be the true proportion of customers who prefer the new design. **B1**

Step 2. Since the researcher is testing for an *increase*, this is a one-tailed test. **B1**

Step 3. Write down H_0 and H_1 using this notation:

$$H_0 : p = 0.3, \quad H_1 : p > 0.3$$

B1

Answer $H_0: p = 0.3$, $H_1: p > 0.3$ (one-tailed test).

EXAM TIP

Defining the parameter in words (“let p be...”) is often its own mark and is frequently skipped by students eager to jump to the symbols. Also remember that a conclusion must always be written in the context of the original claim (“there is sufficient evidence that the proportion has increased”), never as a bare statistical statement like “reject H_0 ” with nothing else.

7.2 Hypothesis testing for a binomial proportion

Concept

When the parameter being tested is a probability of success p , the test statistic is the observed number of successes, X , which follows a binomial distribution under H_0 . The p-value is the probability of obtaining a result at least as extreme as the one observed, calculated using the binomial distribution with H_0 's value of p , then compared to the significance level.

Key vocabulary

Term	Meaning
Test statistic	The observed value from the sample used to make the decision (here, X , the number of successes)
“At least as extreme”	Values as far from, or further from, the expected outcome than the one observed, in the direction of H_1

Procedure

1. Set up $H_0: p = p_0$ and H_1 (one- or two-tailed) as in 7.1, and identify n and the observed number of successes, x .
2. Assuming H_0 is true, $X \sim B(n, p_0)$. Calculate $P(X \geq x)$ if testing for an increase, or $P(X \leq x)$ if testing for a decrease, using the binomial cumulative distribution.
3. Compare this p-value to the significance level: if the p-value is less than the significance level, reject H_0 ; otherwise, there is insufficient evidence to reject H_0 .
4. Write the conclusion in context, referring to the original claim.

Worked example

EXAMPLE 7.2

A company claims that 40% of its customers are members of its loyalty scheme. In a random sample of 20 customers, 13 were found to be members. Test, at the 5% significance level, whether the true proportion of members is greater than 40%.

Step 1. Set up the hypotheses. Let p be the true proportion of loyalty scheme members. **B1**

$$H_0 : p = 0.4, \quad H_1 : p > 0.4$$

Step 2. Under H_0 , $X \sim B(20, 0.4)$. Since this is testing for an increase, find $P(X \geq 13)$. **M1**

$$P(X \geq 13) = 1 - P(X \leq 12) = 1 - 0.9790 = 0.0210 \text{ (3 s. f.)}$$

A1

Step 3. Compare to the significance level: $0.0210 < 0.05$, so the result is significant. **M1**

Answer Since $0.0210 < 0.05$, there is sufficient evidence at the 5% level to reject H_0 and conclude that the true proportion of loyalty scheme members is greater than 40%.

EXAM TIP

A very common source of lost marks is finding $P(X = 13)$ instead of $P(X \geq 13)$ — remember that “at least as extreme” always means a cumulative probability including every more extreme value in the direction of H_1 , never the probability of the single observed value alone.

7.3 Hypothesis testing for the mean of a Normal distribution

Concept

When testing a claim about a population mean μ , we use a sample mean, $\bar{X}$, calculated from the data. The key fact is that if the underlying population is Normal with known variance σ^2 , then $\bar{X}$ itself follows a Normal distribution with the same mean μ , but with a smaller variance, σ^2/n . This means $\bar{X}$ can be standardised and tested in exactly the same way as an individual Normal value, using this reduced variance.

Key vocabulary

Term	Meaning
Sampling distribution of the mean	The distribution of $\bar{X}$ across repeated samples of size n

Key formula

KEY FORMULA

Under H_0 , the sample mean has this distribution:

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

so the test statistic is standardised as:

$$Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

Procedure

1. Set up $H_0: \mu = \mu_0$ and H_1 (one- or two-tailed) based on the claim being tested.
2. Calculate the test statistic z using the sample mean, the hypothesised mean μ_0 , the known population standard deviation σ , and the sample size n .
3. Compare z to the critical value for the given significance level (from Normal tables), or convert z to a p -value and compare to the significance level directly.
4. State the conclusion in context.

Worked example

EXAMPLE 7.3

A machine is set to fill bottles with a mean volume of 500 ml, and it is known that the volumes are Normally distributed with standard deviation 4 ml. A sample of 25 bottles has a mean volume of 498 ml. Test, at the 5% significance level, whether the mean volume has decreased.

Step 1. Set up the hypotheses. **B1**

$$H_0 : \mu = 500, \quad H_1 : \mu < 500$$

Step 2. Calculate the test statistic. **M1**

$$z = \frac{498 - 500}{4/\sqrt{25}} = \frac{-2}{0.8} = -2.5$$

A1

Step 3. For a one-tailed test at 5%, the critical value from tables is -1.6449 . Since $-2.5 < -1.6449$, the test statistic lies in the critical region. **M1**

Answer Since $z = -2.5$ is more extreme than the critical value of -1.6449 , there is sufficient evidence at the 5% level to conclude that the mean volume has decreased.

EXAM TIP

Students often forget to divide σ by $\sqrt{n}$ in the denominator, effectively testing a single observation rather than a sample mean — this dramatically weakens the test's sensitivity and gives the wrong conclusion. Remember that the variance of the sample mean shrinks as the sample size grows, which is exactly what makes larger samples more powerful at detecting a real change.

7.4 Hypothesis testing for zero correlation

Concept

The sample PMCC, r , measures the strength of linear correlation seen in one particular sample. This test asks whether r is large enough to give good evidence that the two variables are genuinely correlated in the whole population, rather than the observed correlation being due to chance in this one sample. The null hypothesis is always that the population correlation coefficient, ρ (rho), is zero, meaning no linear correlation exists.

Unlike the tests in 7.2 and 7.3, this test does not use a test statistic formula. Instead, the sample PMCC is compared directly with a **critical value** read from a table, for the given sample size and significance level.

Key vocabulary

Term	Meaning
ρ (rho)	The population product moment correlation coefficient (as opposed to r , the sample value)

Term	Meaning
Critical value	The threshold value of r , from tables, beyond which H_0 is rejected

Procedure

1. Set up the hypotheses: $H_0: \rho = 0$, and $H_1: \rho > 0$, $\rho < 0$, or $\rho \neq 0$ depending on whether the test is one- or two-tailed.
2. Calculate the sample PMCC, r (or use the value given in the question).
3. Look up the critical value for the given sample size, n , and significance level (halved for a two-tailed test) from PMCC critical value tables.
4. Compare $|r|$ to the critical value: if $|r|$ exceeds the critical value (in the direction of H_1), reject H_0 .
5. State the conclusion in context, referring to the two variables involved.

Worked example

EXAMPLE 7.4

A researcher calculates the PMCC between hours of revision and exam score for a sample of 12 students as $r = 0.62$. Test, at the 5% significance level, whether there is evidence of positive correlation between the two variables in the population.

Step 1. Set up the hypotheses. **B1**

$$H_0 : \rho = 0, \quad H_1 : \rho > 0$$

Step 2. This is a one-tailed test with $n = 12$ at the 5% level. The critical value from tables is 0.4973. **M1**

Step 3. Compare: $r = 0.62 > 0.4973$, so r lies in the critical region. **M1**

Answer Since 0.62 exceeds the critical value of 0.4973, there is sufficient evidence at the 5% level to conclude that there is positive correlation between hours of revision and exam score in the population.

EXAM TIP

A common mistake is looking up the two-tailed critical value for a one-tailed test (or vice versa) — tables are usually laid out with separate columns, and you must match the column to the correct tail based on H_1 , not just to the stated significance level. Always double check which column the question's significance level and tail-type correspond to before comparing.

Chapter summary

Idea	Key point
H_0 and H_1	H_0 is the assumed value; H_1 states the direction (or just inequality) of the suspected change

Idea	Key point
Significance level	The threshold p-value for rejecting H_0 (commonly 5% or 1%)
Binomial test	Find $P(X \geq x)$ or $P(X \leq x)$ under H_0 , compare to the significance level
Normal mean test	Standardise $\bar{X}$ using $\sigma/\sqrt{n}$, compare z to the critical value
PMCC test	Compare $ r $ to a critical value from tables for the given n and significance level
Conclusion	Always written in the context of the original claim, never as a bare statistical statement

Common exam mistakes

- Finding $P(X = x)$ instead of a cumulative “at least as extreme” probability.
- Forgetting to divide the population standard deviation by $\sqrt{n}$ when testing a sample mean.
- Using a two-tailed critical value for a one-tailed test, or vice versa.
- Writing a conclusion as “reject H_0 ” without referring back to the original context of the claim.

You should now be able to

- Set up null and alternative hypotheses for a given claim, including choosing one- vs two-tailed tests.
- Carry out a hypothesis test for a binomial proportion using the p-value or critical region method.
- Carry out a hypothesis test for the mean of a Normal distribution with known variance.
- Carry out a hypothesis test for zero correlation using the PMCC and critical value tables.
- Write conclusions to hypothesis tests clearly, in the context of the original claim.